



**SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY**
INFORMATICS
TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

**DRAFT - Addressing Volumetric Bias in
Multi-Instance Semantic Segmentation
Using Voronoi Tesselation**

Sebastian Oßner



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Adressierung volumetrischer Befangenheit in
Multi-Instanz-Semantischer Segmentierung
durch Voronoi-Tessellation

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Disclaimer

I confirm that this master's thesis is my own work and I have documented all sources and material used.

Munich, 23.04.2026

Sebastian Oßner

Abstract

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1 Introduction

 **Easy intro**

1.1 Addressing the Instance Imbalance Problem

Instance imbalance refers to the issue of certain segmentation instances in biomedical images to be larger than others, causing a bias in common segmentation losses that will lead to smaller instances being harder to segment. This issue is especially problematic when the pathology to be segmented shows no correlation between size and clinical relevance. This makes the segmentation of smaller instances a high-stakes issue. (M. F. Rachmadi et al. 2024; Florian Kofler, Shit, et al. 2023; Zhang et al. 2021; F. Rachmadi et al. 2024)

1.2 Voronoi Tesselation

Voronoi diagrams are a fundamental building block of the approach outlined in this thesis. They define a partition of a space into regions. In the simple case of the euclidean plane, a finite set of points $\{p_1, p_2, \dots, p_n\}, n \in \mathbb{N}$ must be given. Each point $p_i, i \in [n]$ is assigned a region R_i . All other points in the plane are assigned the region that minimizes some distance function d :

$$R_k = \{x \in X \mid d(x, P_k) \leq d(x, P_j), \forall j \neq k\} \quad (1)$$

In our case, there are  **Formalize this and describe the 3D case and instances instead of points**

1.3 Connected Components

 **Connectivity, formal definition and clinical principal impact**

2 Related Work

2.1 A Taxonomy of Losses

 **Adapt and update figure from** (Ma et al. 2021; Zhang et al. 2021; F. Rachmadi et al. 2024; M. F. Rachmadi et al. 2024; Florian Kofler, Shit, et al. 2023; Bouteille et al. 2025)

3 Methodology

3.1 Metrics and Fidelity of Multi-Instance Segmantation

Datasets

The theoretical basis of this thesis was evaluated against multiple datasets that span dimensionality (2D as well as 3D), modality (CTs, electron microscopy) and various inherent anatmoical features and pathologies that result in varied segmentation instance properties.

Due to the diverse nature of the underlying data, it was imperative to gather dataset metrics that encapsulate these varied instance properties not only to properly evaluate our fundamental hypotheses, but also to deal with noise and errors during training. In order to investigate why certain approaches worked better than others, concrete information on the size, morphology, distribution, and number of instances must be reported and taken into consideration before conclusions can be drawn. This subsection contains a description of the datasets used, their properties and calculated metrics as well as a comprehensive overview on the estimation of their fidelity that could serve as a basis for future research in multi-instance segmentation.

Since this thesis concerns itself with bianry semantic segmentation, all datasets can be abstracted into their constituend components as follows: An image of shape $h * w$ ($h * w * d$ in the case of 3D) and a binarized label file of the same shape as its corresponding image. The label file is therefore partitioned into background pixels/voxels and $n \geq 0$ foreground pixels/voxels. The n foreground pixels/voxels then undergo a process of labeling based on their local neighborhood, in which they are split into $m \geq 0$ foreground instances (typically $n \gg m$). In 2D, this is based on 8-connectivity while 3D instances are labeled using the higher-dimensional equivalent of 26-connectivity.

The resulting instances have inherent properties that are important to examine both when formulating hypotheses as well as investigating noise and segmentation errors. As these properties are of such importance, prior works in the fields of [Medical Imaging Quality Assessment \(MIQA\)](#) and [Topological Data Analysis \(TDA\)](#) have provided comprehensive frameworks for identifying the properties of segmentation data and how these impact performance reporting ([Florian Kofler, Möller, et al. 2023](#); [Maier-Hein et al. 2022](#)). While these works place substantial focus on the selection and calculation of quantitative segmentation metrics, this section provides a more holistic interpretation of multi-instance dataset attributes that shall aid in the interpretation of results.

 **Description of removal of small annotation errors and what theoretical effects these instances would have and a link to the evaluation of intact vs cleaned datasets**

3.1.1 Brain Metastases

 **Split description, instance distribution, sizes and variance, etc.**

3.1.2 White Matter Hyperintensities

 **Split description, instance distribution, sizes and variance, etc.**

3.1.3 Platelets

 **Split description, instance distribution, sizes and variance, etc.**

3.1.4 Mitochondria

 **Split description, instance distribution, sizes and variance, etc.**

3.2 Segmentation Evaluation Metrics

Many works have previously discussed the importance of the choice of metrics and the need to adapt to the specific task at hand, (Maier-Hein et al. 2022) have provided concrete guidance in the choice of instance-wise metrics in segmentation problems and (Florian Kofler, Möller, et al. 2023) provide a tool to calculate these metrics.

Additionally, (Jaus et al. 2025) proposed an additional family of metrics that are of particular interest to us since they also use voronoi tessellation and aggregate metrics on each region separately to address instance volume imbalance during evaluation.

This subsection will provide a comprehensive overview of the metrics of interest, the rationales behind this choice and supplementary information on how predicted segmentations were evaluated.

 **Relatively metrics-agnostic implementation: No checkpointing, training to completion, mirror sentiment from nnUNet**

3.2.1 Global Metrics

Standard Dice, F2

3.2.2 Instance-wise Metrics

Instance F1, Instance Dice, CCDice, Instance Recall by volume

3.3 Loss Formulation and Weighted Combination

 **Tested Losses, Global vs. Local splits and weight distribution and normalization as potential shortcoming of previous studies** Why does everyone combine local and global losses? Why didn't anyone account for potential absolute value inconsistency when doing so?

3.4 Weight Maps

 **A description and formulation of tested weight maps and their impact on segmentation results** A special note of instance cleaning and weight maps inverse weighting as previously implemented. Voronoi-based maps as novel and precomputed due to image patching

3.5 Model Architecture

3.5.1 Precomputation and Image Patching

 **Instance information, voronoi maps, and weight maps can be precomputed and patched alongside images and labels.** This can lead to artifacts in train patches that are devoid of instances, but are still tessellated as a region and therefore punish FPs more harshly than in LTLC.

4 Results

 An instance-based analysis of segmentation, analyse instance metadata like morphology and voronoi regions in results  Dataset noise and coeff. of covariance per dataset and how the random seed in deterministic training influences results  Global vs instance vs mixed with a focus on global vs mixed due to earlier established literature  Global vs. local weight distribution  Analysis of runtimes and relative efficiency  Weight maps  The effect of false instance removal on partitioning and weight maps in 2D  Adaptive Weighting

5 Discussion

6 Conclusion

Appendix A: Supplementary Material

– Supplementary Material –

 **Your appendix content goes here**

Glossary

MIQA – Medical Imaging Quality Assessment [3](#)

TDA – Topological Data Analysis [3](#)

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